

NGAME Terminology Glossary

Three-Layer Model for Churn, Detection, and Triage

Status: Draft for journal manuscript

Purpose: Replace confusion-matrix labels (TP / FP / FN / TN) on the Jaccard 2x2 with precise, layer-specific terms.

Audience: Co-authors, reviewers, and readers of the NGAME design-science evaluation.

Why This Glossary Exists

NGAME uses “positive” and “negative” in three distinct senses. Conflating them invites reviewer criticism and misstates what the artifact does. This glossary assigns **one vocabulary per layer** and retires TP/FP/FN/TN on the day-over-day record-overlap matrix (Layer 1). Standard confusion-matrix language is reserved for **controlled evaluation against ground truth** (Layer 2). FRP-facing outcomes use **triage vocabulary** that does not imply autonomous fraud verdicts (Layer 3).

Layer Overview

Layer	Question answered	Primary outputs	Ground truth
1 — Structural	How stable is each transaction-type population from yesterday to today?	CPI, α (addition rate), δ (deletion rate), wCPI	None required; descriptive set overlap
2 — Detection	Which ϕ dimensions deviate enough from the trained baseline to rank in the top- k alert set?	z-scores, HIGH / MEDIUM / LOW bands, top- k rankings	Injected or known anomalies (evaluation only)
3 — Triage	What should the Financially Responsible Person (FRP) review, and what did review conclude?	Management warnings, overall risk, dashboard artifacts	FRP professional judgment after review

Table 1 — Layer 1: Day-over-Day Record Overlap (Jaccard / CPI)

Replaces the Jaccard 2x2 labels TP, FP, FN, TN in the journal version.

Rows: presence in **Yesterday's** snapshot (Y). Columns: presence in **Today's** snapshot (T).

Cell	Preferred term	Symbol	Set notation	Role in CPI	Interpretation
Upper-left	Retained	a	$Y \cap T$	Numerator of CPI	Record URI present in both snapshots; continuity across days
Upper-right	Removed	b	$Y \setminus T$	Contributes to δ (deletion rate)	Record existed yesterday, absent today; churn or deletion
Lower-left	Added	c	$T \setminus Y$	Contributes to α (addition rate)	Record absent yesterday, present today; new entry
Lower-right	Excluded	—	(not counted)	Not in formula	Absent both days; universe of non-occurring records is undefined

Core formulas (Layer 1 only):

Quantity	Formula	Range
CPI (Churn Persistence Index)	$CPI = a / n$, where $n = a + b + c$	[0, 1]
Addition rate (α)	$\alpha = c / n$	[0, 1]
Deletion rate (δ)	$\delta = b / n$	[0, 1]
Partition identity	$CPI + \alpha + \delta = 1$	—

Edge convention: When $n = 0$ (no records of type k in either snapshot), $CPI = 1.0$ — absence is treated as stable.

Dollar-weighted variant: Same structure with dollar weights w_i instead of URI counts → wCPI, wAdditionRate, wDeletionRate.

Retired Layer 1 labels (do not use in journal prose)

Retired label	Was used to mean	Use instead
TP (True Positive)	Retained	Retained (a)
FP (False Positive)	Added	Added (c)
FN (False Negative)	Removed	Removed (b)
TN (True Negative)	Excluded	Excluded (state explicitly that this cell is omitted from Jaccard)

Note: “Added” is not a “false” outcome. A legitimate new vendor is **Added** at Layer 1 and may still yield a LOW z-score at Layer 2 if consistent with the trained baseline.

Table 2 — Layer 2: Detector Evaluation Metrics (Ground-Truth Comparison)

Use standard IR / confusion-matrix terms only here, when comparing NGAME’s top- k flagged ϕ dimensions to known or injected anomalies.

Detection task (evaluation): Given an N -element daily ϕ (CPI) array, rank dimensions by $|z|$ (or composite alarm) and treat the top- k as the alert set ($k = 3$ in current instantiation; $N = 17\text{--}18$).

Term	Symbol	Definition	Formula
True Positive (eval.)	TP	Flagged dimension where an anomaly was knowingly introduced or present in ground truth	—
False Positive (eval.)	FP	Flagged dimension with only normal variation relative to ground truth	—

Term	Symbol	Definition	Formula
False Negative (eval.)	FN	Ground-truth anomaly dimension not in the top- k alert set	—
True Negative (eval.)	TN	Normal-variation dimension not flagged	—
Precision	—	Share of flagged dimensions that match ground truth	$TP / (TP + FP)$
Recall (top-k)	—	Share of anomalous dimensions captured in top- k	$TP / (TP + FN)$
False Positive Rate	FPR	Share of normal dimensions incorrectly flagged	$FP / (FP + TN)$

Scope limits (state in manuscript):

- Metrics apply to **transaction-type dimension ranking**, not dollar amount of any single transaction.
- Formal element-level precision/recall requires per-pattern logging of top- k rankings against ground truth; pipeline pass/fail alone is insufficient.
- Design intent: triage favors **recall over precision** given ACFE-style detection lag without continuous monitoring.

Layer 2 outputs (operational, pre-FRP):

Term	Definition
z-score	$z_k = (CPI_k(\text{today}) - \mu_k) / \sigma_k$; signed deviation from trained mean
Deviation band	$ z > 3.0 \rightarrow \text{HIGH}$; $ z > 2.0 \rightarrow \text{MEDIUM}$; else LOW
Composite alarm	Fires when count-based or dollar-weighted signal reaches MEDIUM or HIGH
Top-k alert set	The k ϕ dimensions with largest $ z $ (or equivalent ranking rule) on a given day

Table 3 — Layer 3: FRP Triage Outcomes (Operational)

NGAME does not accuse anyone of fraud. A warning means activity is statistically unusual relative to the organization's 30-day baseline; the FRP adjudicates.

Term	Definition	Relation to Layer 2
Statistically warranted alert	z or composite alarm exceeds threshold; signal is real relative to trained μ , σ	May correspond to FP (eval.) if ground truth is "normal business change"
Corroborated alert	FRP review finds material concern: fraud, control weakness, or required correction	Aligns with TP (eval.) when ground truth exists; broader in production
Benign resolution	Alert correctly reflected baseline deviation; FRP explains as legitimate (e.g., new GL accounts, seasonal vendor onboarding)	Often FP (eval.) under fraud ground truth; not a system failure under triage design
Missed event	Material misappropriation occurred but did not surface in warnings or top- k	FN (eval.) when ground truth is known

Terms to avoid at Layer 3

Avoid	Reason	Prefer
False positive (unqualified)	Implies detector error; conflates with Layer 2	Benign resolution or statistically warranted, benign on review
True positive (unqualified)	Implies fraud confirmed by the system	Corroborated alert (after FRP review)
Unreliable warning	Suggests broken instrumentation	Benign resolution or spurious for fraud purposes, valid for triage
Reliable (for warnings)	Collides with "reliable μ and σ estimates" (Layer 1 training)	Corroborated, warranted, or actionable after review

Table 4 — Cross-Layer Mapping (Illustrative)

One day’s Vendors dimension can illustrate how layers diverge:

Layer	Example reading
1 — Structural	High Added and Removed counts → low CPI (churn)
2 — Detection	$ z > 3$ on Vendors → HIGH; Vendors in top- k
3 — Triage	FRP reviews Audit Log → benign resolution (legitimate vendor cleanup) or corroborated alert (shell-vendor pattern)

Chart of Accounts example (field deployment): Layer 2 correctly flags HIGH deviation; Layer 3 **benign resolution** after FRP confirms legitimate new GL accounts. Not an eval. FN; triage behaved as designed.

Table 5 — Notation Quick Reference

Symbol	Meaning
Y, T	Yesterday and Today ontology snapshots for a transaction type
a, b, c, n	Retained, removed, added, and union count ($n = a + b + c$)
ϕ_k, CPI_k	Churn Persistence Index for transaction type k
μ_k, σ_k	Trained mean and standard deviation of CPI_k (30-day baseline)
$wCPI_k$	Dollar-weighted CPI for type k
$SCPI_k(t, L)$	Sequence CPI: geometric mean of CPI_k over window L
z_k	Standardized deviation: $(CPI_k(\text{today}) - \mu_k) / \sigma_k$
k (alert)	Number of top-ranked anomalous dimensions in evaluation (default 3)
FRP	Financially Responsible Person — designated reviewer

Suggested Manuscript Footnote (optional paste-in)

Terminology: Day-over-day record overlap (retained, added, removed) describes structural churn only and is not labeled with confusion-matrix terms. True and false positives,

precision, and recall apply solely to evaluation of top- k dimension ranking against known or injected anomalies. Management warnings indicate statistical departure from an organization-specific baseline; the FRP classifies outcomes as corroborated, benign on review, or missed—NGAME does not render autonomous fraud verdicts.

Draft aligned with NGAME CPI analysis (`ngame_cpi_analysis_agent.py`), churn comparison thresholds (`ngame_churn_comparison_agent.py`), FRP Operations Guide, and DSRM Evaluation Draft §5.2.

[QLMarkdown](#), version 1.0.24 (50).

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